

Computational Hydrodynamic Modeling of Coastal Brine Dispersion

Selected computational modeling and results excerpt from M.Sc. thesis

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Scope note. This document is a selected and condensed English adaptation of the computational-modeling sections of my M.Sc. thesis. The equations, numerical settings, sensitivity tests, validation metrics, maps, and simulation results are taken from the original thesis. No new simulations or results are introduced.

Problem and computational setting

The research modeled the transport and dilution of negatively buoyant brine discharged from a seawater reverse-osmosis plant into Chabahar Bay. The computational task was to resolve a three-dimensional coastal flow field, couple it to salinity transport, test numerical resolution, calibrate the hydrodynamics against observations, and then use the validated model to compare discharge scenarios.



Figure 1. Chabahar Bay and the desalination-plant location shown in the original thesis (Google Earth imagery reproduced from the thesis).

1. Governing equations and coupled flow-transport model

The hydrodynamic component was implemented with MIKE 3 Flow Model FM. The thesis describes the model as a three-dimensional incompressible Navier-Stokes system using the Boussinesq approximation and hydrostatic pressure assumption. Continuity and horizontal momentum equations are coupled to temperature and salinity transport, while unresolved mixing is represented through an eddy-viscosity closure.

Continuity

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = S \quad (1)$$

x-momentum

$$\begin{aligned} \frac{\partial u}{\partial t} + \frac{\partial u^2}{\partial x} + \frac{\partial(vu)}{\partial y} + \frac{\partial(wu)}{\partial z} = & f_v - g \frac{\partial \eta}{\partial x} - \frac{1}{\rho_0} \frac{\partial p_a}{\partial x} - \frac{g}{\rho_0} \int_z^\eta \frac{\partial \rho}{\partial x} dz \\ & - \frac{1}{\rho_0 h} \left(\frac{\partial s_{xx}}{\partial x} + \frac{\partial s_{xy}}{\partial y} \right) + F_u + \frac{\partial}{\partial z} \left(\nu_t \frac{\partial u}{\partial z} \right) + u_s S \end{aligned} \quad (2)$$

y-momentum

$$\begin{aligned} \frac{\partial v}{\partial t} + \frac{\partial v^2}{\partial y} + \frac{\partial(uv)}{\partial x} + \frac{\partial(wv)}{\partial z} = & -f_u - g \frac{\partial \eta}{\partial y} - \frac{1}{\rho_0} \frac{\partial p_a}{\partial y} - \frac{g}{\rho_0} \int_z^\eta \frac{\partial \rho}{\partial y} dz \\ & - \frac{1}{\rho_0 h} \left(\frac{\partial s_{yx}}{\partial x} + \frac{\partial s_{yy}}{\partial y} \right) + F_v + \frac{\partial}{\partial z} \left(\nu_t \frac{\partial v}{\partial z} \right) + v_s S \end{aligned} \quad (3)$$

Figure 2. Continuity and horizontal momentum equations, typeset from the governing equations reproduced in the original thesis methodology.

Physical and numerical variables

Component	Representation in the thesis
Flow field	Three-dimensional velocity and free-surface elevation
Density effects	Baroclinic formulation with salinity- and temperature-dependent density
Scalar transport	Advection-diffusion equations for salinity and temperature
Mixing / turbulence	Vertical eddy viscosity; Smagorinsky formulation used in calibration
External forcing	Tides, wind, Coriolis forcing, bathymetry, and discharge source terms
Discharge behavior	Negatively buoyant jet/plume characterized using densimetric Froude number and dilution

The thesis also defines the scalar transport equations explicitly and uses the modeled salinity field as the principal output for plume analysis. This makes the problem a coupled fluid-flow and transport simulation rather than a stand-alone empirical dispersion estimate.

2. Numerical setup, resolution sensitivity, and boundary conditions

The computational domain was built from coastal bathymetry and an open boundary toward the Gulf of Oman. Time-varying tidal water level was prescribed at the open boundary; wind forcing was taken from ECMWF data, and the vertical coordinate used sigma layers.

Key calibration setup

Parameter	Value / formulation
Time-step interval	3600 s
Time integration	Low-order, fast algorithm
Spatial discretization	Low-order, fast algorithm
Vertical system	Equidistant sigma layers
Density type	Baroclinic
Eddy-viscosity formulation	Smagorinsky
Open-boundary condition	Specified, time-varying water level

Mesh-size sensitivity

Mesh area (deg ²)	Run time for 1 model year (h)	Velocity change vs. finest mesh
0.00001	48	0%
0.00005	12	4%
0.0005	6	5%
0.001	5	8%
0.01	3	15%

Vertical-layer sensitivity

Sigma layers	Run time for 1 model year (h)	Velocity change vs. 7 layers
3	3	1.4%
5	4	0.1%
7	6	0%

Resolution choice. The 0.0005 deg² mesh was selected because the modeled current velocity changed by about 5% relative to the finest mesh while the one-year run time decreased from 48 h to 6 h. Three sigma layers were retained because the velocity response differed only slightly from the seven-layer reference while computational time was approximately halved.

3. Calibration and quantitative validation

The hydrodynamic model was calibrated against measured water-level data at Chabahar Port. The thesis reports one-year model runs in which bed roughness and the Smagorinsky eddy-viscosity parameter were varied. The selected calibration values were a bed-roughness height of 0.01 and a Smagorinsky coefficient of 0.28.

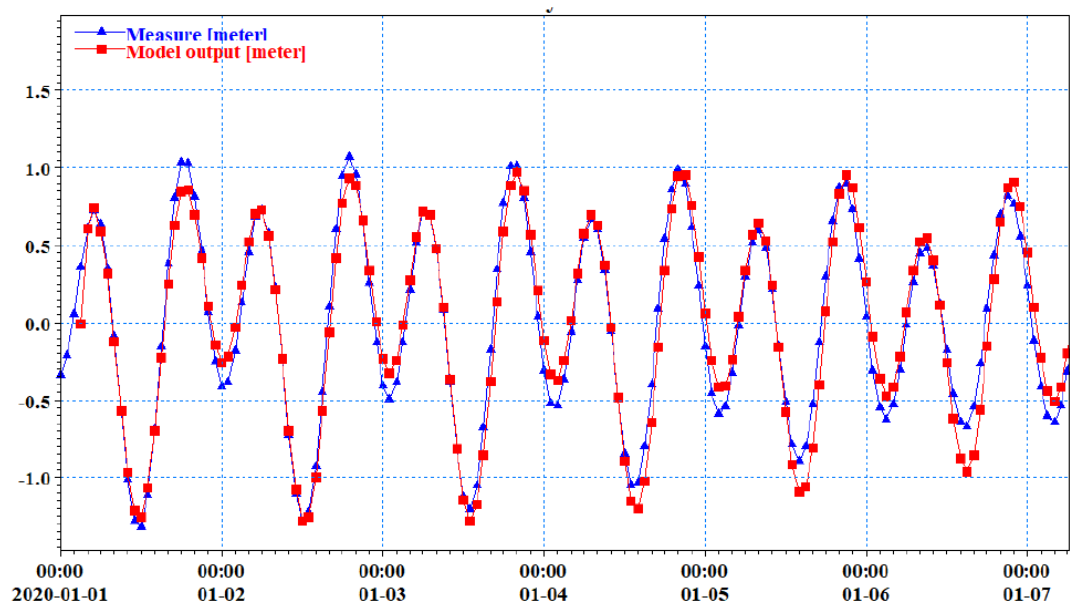


Figure 3. Measured and simulated water level at Chabahar Port during the first week of January 2020.

Validation metric	Reported value
Root Mean Square Deviation (RMSD)	0.22 m
Pearson correlation coefficient (r)	0.92
Nash-Sutcliffe Efficiency (NSE)	0.85

Model-use criterion. Based on the close agreement in water-level timing and amplitude and the reported performance metrics, the thesis treated the calibrated hydrodynamic model as sufficiently accurate for the subsequent plume-scenario simulations.

4. Representative simulation result: current outfall

After calibration, the model was used to resolve the salinity plume under combinations of discharge rate, tidal condition, wind speed, and wind direction. The figure below is a representative high-discharge case from scenario 3 with a 10 m/s wind from 225°, showing both the plan-view salinity field and a vertical profile through the plume.

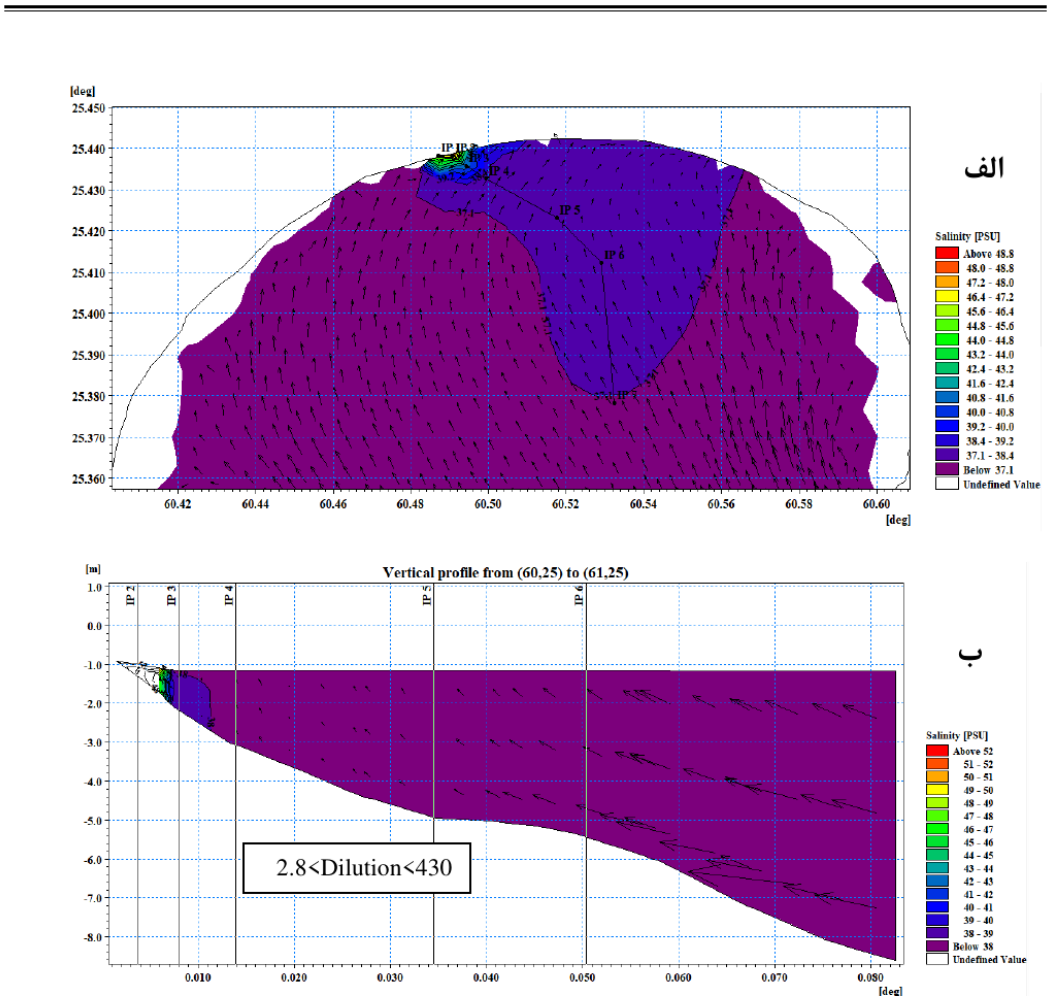


Figure 4. Representative three-dimensional salinity-plume result for the current discharge configuration (scenario 3, wind 10 m/s from 225°), reproduced from the thesis.

Result reported in the thesis. For the current outfall, the thesis found that the highest salinity during ebb conditions exceeded the 41 PSU environmental threshold in the 200 m assessment zone. Under 225° wind conditions, scenario 3 reached approximately 52 PSU within 200 m and the plume extended several kilometers from the discharge point. The vertical profile also shows the dense plume remaining close to the bed as it is transported away from the source.

5. Scenario-based redesign: alternative discharge point

The validated model was then used as a design tool. Two alternative discharge locations were evaluated, and the second point was placed farther offshore and deeper than the existing surface discharge. The selected configuration used a single-port outlet approximately 90 m from the shoreline, at about 1.2 m depth, with a 60° vertical discharge angle.

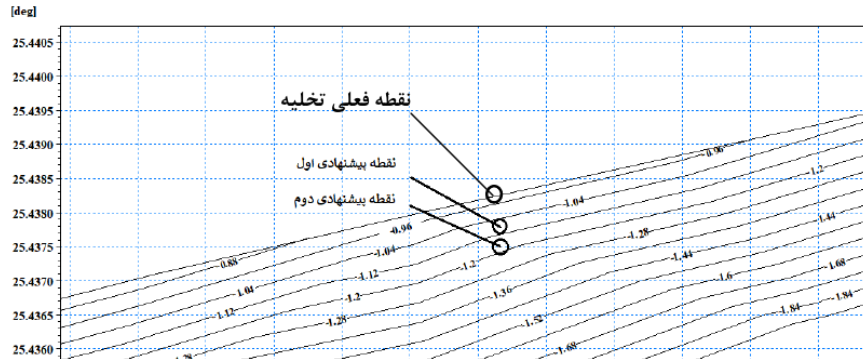


Figure 5. Existing discharge point and two proposed alternative locations from the thesis model domain.

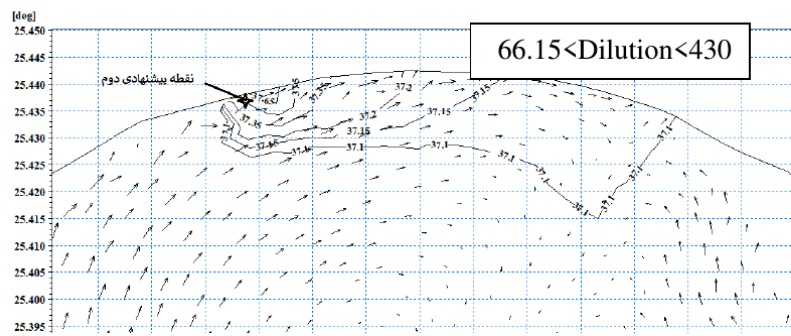


Figure 6. Salinity contours and current vectors for proposed point 2 under a 10 m/s wind from 225°, reproduced from the thesis.

Design outcome. For proposed point 2, the thesis reports a maximum salinity of about 38 PSU within the 200 m assessment zone in the evaluated scenario, compared with substantially higher values for the current configuration. This scenario was therefore selected as the preferred discharge location for the third project phase.

What this excerpt demonstrates

- Formulation and use of a coupled three-dimensional coastal flow and scalar-transport model.
- Numerical-resolution testing with explicit accuracy-versus-computational-cost trade-offs.
- Calibration and validation against measured field data using quantitative performance metrics.
- Interpretation of multidimensional simulation outputs, including plan-view fields, velocity vectors, and vertical profiles.
- Use of the validated model for scenario comparison and a concrete engineering design decision.

Original thesis title: Investigation of Brine Dispersion from the Chabahar Desalination Plant into the Sea Using Numerical Modeling (M.Sc. thesis, University of Tehran, 2022).